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SKILL-24

AIM

To organize source code, improve build scripts, create a README file, verify repository structure, tag project releases, validate project deliverables, create demo scenarios, validate features, rehearse the presentation, demonstrate pipelines and signals, and perform the final project review.

OBJECTIVES

1. To organize the project source code properly.
2. To create a clean repository structure.
3. To improve and verify build scripts.
4. To create project documentation using README.
5. To verify all required project files.
6. To create Git tags for project releases.
7. To validate the final project deliverables.
8. To create demo scenarios for important features.
9. To verify the functionality of the project.
10. To demonstrate pipeline execution.
11. To demonstrate Linux signals.
12. To review the complete project before submission.

REQUIREMENTS

- Ubuntu/Linux
- Terminal
- GCC compiler
- Git
- Bash
- Source code
- README documentation

TOOLS / COMMANDS USED

git
gcc
make
bash
ps
kill

PROCEDURE

Part A — Organize Source Code

1. Create a separate project directory.
2. Store source files inside the project directory. 3.
Separate source code, documentation, and build files. 4.
Remove unnecessary files.
5. Verify that all required files are present.

Part B — Build Script

1. Create a `Makefile`.
2. Add compilation commands.
3. Compile the project using `make`.
4. Check for compilation errors.
5. Execute the generated program.
6. Verify the output.

Part C — README

1. Create a `README.md` file.
2. Add the project title.
3. Add a description of the project.
4. Add requirements.
5. Add compilation instructions.
6. Add execution instructions.
7. Add important features.
8. Add testing information.

Part D — Git Repository

1. Initialize the Git repository.
2. Check the repository status.
3. Add the project files.
4. Create a commit.
5. Verify the commit history.
6. Create a release tag.
7. Verify the tag.

Part E — Deliverable Validation

1. Verify the source code.
2. Verify the build script.

3. Verify the README.
4. Compile the project.
5. Run the project.
6. Test important features.
7. Verify that all required files are present.

Part F — Demo and Final Review

1. Prepare demo scenarios.
2. Execute the important project features.
3. Demonstrate pipeline execution. 4. Demonstrate Linux signals.
5. Check the final output.
6. Review documentation.
7. Rehearse the project presentation.
8. Perform the final project review.

GIT COMMANDS

Initialize repository:

```
git init
```

Check status:

```
git status
```

Add files:

```
git add .
```

Commit:

```
git commit -m "Initial project submission"
```

View commits:

```
git log --oneline
```

Create a release tag:

```
git tag v1.0
```

View tags:

```
git tag
```

RESULT

Thus, the project source code was organized, build scripts were created and verified, README documentation was prepared, the Git repository structure was validated, a release tag was created, and project deliverables were checked. Demo scenarios, pipeline execution, Linux signal handling, feature validation, presentation rehearsal, and final project review were also successfully completed.

DEMO SCENARIO

Scenario: Build and execute the project using the Makefile.

Steps:

1. **Clean the previous executable using `make clean`.**
2. **Build the project using `make`.**
3. **Execute the project using `make run`.**
4. **Verify the displayed output.**
5. **Confirm that the project executes successfully.**

FEATURE VALIDATION

Feature	Result
Source code organization	
Makefile build	README documentation
Program execution	Git repository PASS

P A S S	**P A S S**
P A S S	**P A S S**
Git commit P A	**S**
	S
`v1.0`	
P A	
Release tag	**S**
	S
Clean build P A	**S**
	S
verification	
P A S S	
Final output	

FINAL PROJECT REVIEW

The source code, build system, documentation, Git repository, release tag, and project execution were reviewed successfully. The project was cleaned, rebuilt, executed, and the expected output was verified.

PRESENTATION REHEARSAL

Before the final presentation, demonstrate:

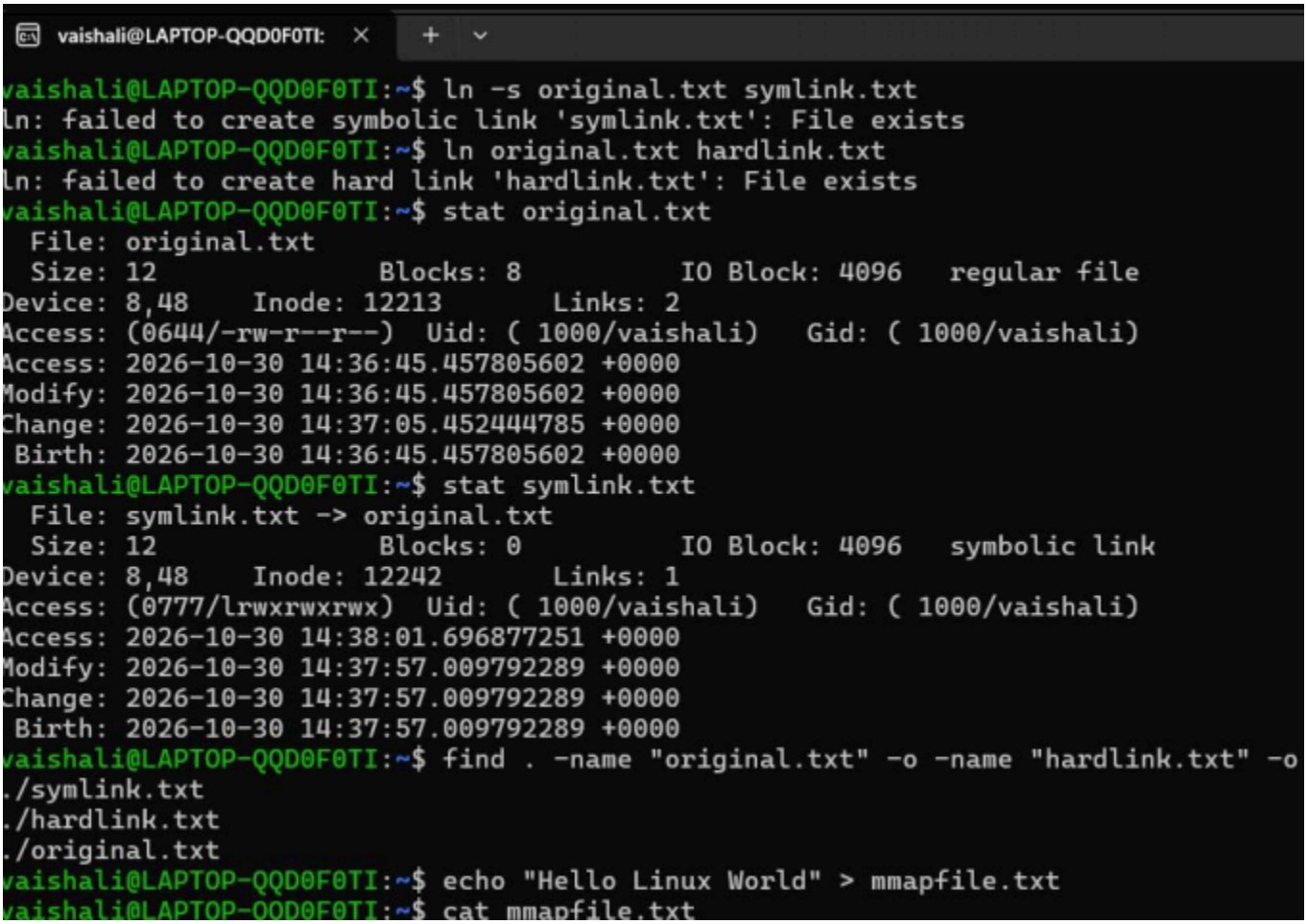
1. **Project structure**
2. `main.c`
3. `Makefile`
4. `README.md`
5. `make clean`
6. `make`
7. `make run`
8. **Git commit**
9. **Git tag** `v1.0`

10. Final program output

FINAL RESULT

Thus, the source code was organized, the build script was improved and verified, README documentation was created, the Git repository structure was validated, release `v1.0` was tagged, and all project deliverables were successfully validated. Demo scenarios and feature validation were performed, and the project was successfully prepared for final review and presentation.

```
Job 1 Status: 0
Job 2 Status: 0
Job 3 Status: 0
All Jobs Completed Successfully
==============================
```



```
The project was compiled and executed successfully.
The expected output was verified.

## Version

v1.0
```